

LAB MANUAL

**II M.Sc COMPUTER SCIENCE
BATCH 2023 – 2024 AND ONWARDS**

**DATA VIRTUALIZATION LAB
23PBICP311**

SEMESTER - III

KONGU ARTS AND SCIENCE COLLEGE(AUTONOMOUS), ERODE – 638107

DEPARTMENT OF COMPUTER SCIENCE (P.G.)

CLASS: II M.Sc COMPUTER SCIENCE

COURSE: CORE PRACTICAL VIII: DATA VIRTUALIZATION LAB

COURSE CODE: 23PBICP311

LIST OF PRACTICALS

1. Create an Excel Worksheet for the monthly sales of a product and also represent the data by using bar chart. (Use www.sheet.com)
2. Prepare a PPT about Data Virtualization with explanations and images. (Use www.show.com).
3. Implementation of Para-Virtualization using VM Ware's Workstation/ Oracle's Virtual Box with different flavours of Linux or windows OS on top of windows 7 or 8.
4. Install Turbo C in guest OS and execute C program.
5. Implement the procedure File Transfer in Client & Server using virtual machine.

1. Create an Excel Worksheet for the monthly sales of a product and also represent the data by using bar chart (Use www.sheet.com)

Aim:

To create a worksheet that keeps tracks of the monthly sales of a product, for easy data analysis and visualization.

Procedure:

Step1: Open Zoho Sheet: Go to zoho.sheet.com and log in.

Step2: Create a New Sheet: Click on “New Spreadsheet.”

Step3: Create and Set Up the Data:

- In Column A, enter the months (e.g., January to December).
- In Column B, enter the corresponding sales figures for each month.

Step4: Format the Data: Highlight the data range (months and sales figures) for better readability.

Step5: Insert Bar Chart:

- Select the data range.
- Go to the “Insert” menu and choose “Chart.”
- Select “Bar Chart” from the chart options.

Step6: Customize the Chart: Adjust titles, labels, and colors as desired.

Step7: Save Your Work: Click “File” and then “Save” to keep your worksheet and chart.

Step8: Click on file → download as → .xlsx to download as excel document

Step9. Click on file → download as → .pdf to download as PDF document

Result:

Thus, we have created worksheet with bar and pie charts to visualize the monthly sales data.

2. Prepare a PPT about Data Virtualization with explanations and images. (Use www.show.com)

Aim:

To present the user about the concept of data virtualization with visual appeals.

Procedure:

To create a presentation worksheet that keeps tracks of the monthly

Step1: Sign In to Zoho Show: Go to zoho.show.com and log in or create an account.

Step2: Create a New Presentation:

- Click on "Create" to start a new presentation.
- Choose a suitable template or start from scratch.

Step3: Outline Your Content:

- Slide 1: Title Slide
- Slide 2: Introduction
- Slide 3: How It Works
- Slide 4: Benefits:
- Slide 5: Use Cases
- Slide 6: Conclusion

Step4: Add Content and Images:

- Insert text, images, and diagrams relevant to each slide.

- Use high-quality visuals to enhance understanding.

Step5: Apply Animations:

- Select elements on a slide, go to the “Animations” tab.
- Choose entrance, emphasis, or exit animations for text and images.
- Adjust animation timing for smooth transitions.

Step6: Review and Finalize:

- Play the slideshow to see animations in action.
- Edit for coherence and clarity.

Step7: Share or Present:

- Save your presentation.
- Use the share option to collaborate or present directly.

Step8: Click on file → download as → .pptx to download as presentation

Step9: Click on file → download as → .pdf to download as PDF document

Result:

Thus, we have created a customized slide with visuals.

Program – 3

Implementation of Para Virtualization

3. a. Implementation of Para-Virtualization using VM Ware’s Workstation with Linux OS

Aim:

To implement para-virtualization using VM Ware’s Workstation with Linux OS

Procedure:

Step 1: Installation of Virtual Machine software

VMware:

1. Open the Google and type VMWare Workstation 16
2. Click the link “Download VMWare Workstation 16”
<https://www.vmware.com/in/products/workstation-player/workstation-player-evaluation.html>
3. Click Try Workstation 16.0 Player for Windows
4. Double click the downloaded file and click “I Accept” in EULA and install the software(EULA-End User License Agreement)

Step 2:

Download the Operating System for the Virtual Machine

Ubuntu

1. Open the Google and type “download ubuntu 16.04.1 amd64 iso”
2. Ubuntu 16.04.1 LTS (Xenial Xerus) - Old Releases <https://old-releases.ubuntu.com/releases/16.04.1/>
3. Select an image → Desktop image → 64-bit PC (AMD64) desktop image to download the Linux OS

Step 3:

Creation of Virtual Machine:

VMware:

1. Open the VMware and click Player → File → New Virtual Machine

2. Select the “I will install the OS later” in the New Virtual Machine Dialog Box.
3. Click Next → Select the Linux OS and select Ubuntu 64 bit. Click Next → set the name of virtual machine and click Next → Next and Finish.
4. Select the created Virtual machine and Click Edit Virtual Machine Settings → Select CD\DVD(SATA) and click Browse and select the ubuntu disk image and save the settings.

Step 4:

Installation of Guest Operating system:

Ubuntu:

1. Click Play Virtual Machine
2. Click Install Ubuntu → continue → Install Now → Continue
3. Set the time zone and click Continue. Set the keyboard layout, language and click continue
4. Set the Name and password and click continue.
5. Click Restart Now after the OS is installed successfully.

Step 5: Shutdown the guest machine to exit from the Virtual machine.

Conclusion:

Para-virtualization of Linux OS using VMWare’s Workstation has been implemented.

3. b. **Implementation of Para-Virtualization using VM Ware’s Workstation with windows 10 OS**

Aim:

To implement para-virtualization using VM Ware’s Workstation with windows 10 OS

Procedure:

Step 1: Installation of Virtual Machine software

VMware:

1. Open the Google and type VMWare Workstation 16
2. Click the link “Download VMWare Workstation 16”
<https://www.vmware.com/in/products/workstation-player/workstation-player-evaluation.html>
3. Click Try Workstation 16.0 Player for Windows
4. Double click the downloaded file and click “I Accept” in EULA and install the software(EULA-End User License Agreement)

Step 2:

Windows

1. Open the Google and type **download windows 10 iso**
2. Click the link in and download the versions 64 or 32 bit file

Step 3:

Creation of Virtual Machine:

VMware:

1. Open the VMware and click Player → File → New Virtual Machine
2. Select the “I will install the OS later” in the New Virtual Machine Dialog Box.
3. Click Next → Select the Linux OS and select Ubuntu 64 bit. Click Next → set the name of virtual machine and click Next → Next and Finish.
4. Select the created Virtual machine and Click Edit Virtual Machine Settings → Select CD\DVD(SATA) and click Browse and select the ubuntu disk image and save the settings.

Step 4:

Installation of Guest Operating system:

Windows:

1. Click Play Virtual Machine

2. After all files are loaded, Click **Install Now** Button.
3. Click **I Accept the terms** in the End User license Agreement.
4. Select the Language and currency and click Next.
5. Select the Drive and Click Next. The Installation process will start.
6. After the installation is complete click Restart Now Button
7. Enter the Username, Password and Time zone.

Step 5: Shutdown the guest machine to exit from the Virtual machine.

Conclusion:

Para-virtualization of windows 7 OS using VMWare's Workstation has been implemented.

3. c. Implementation of Para-Virtualization using Oracle's Virtual Box with Linux OS

Aim:

To implement of Para-Virtualization using Oracle's Virtual Box with Linux OS

Procedure:

Step 1: Installation of Virtual Machine software

VirtualBox:

1. Open Google and type Oracle VirtualBox for Windows
2. Click the link and download the Oracle VirtualBox x86_64.exe
3. After downloading double click the file and click "I Accept" in EULA and install the software

Step 2:

Download the Operating System for the Virtual Machine

Ubuntu

4. Open the Google and type "download ubuntu 16.04.1 amd64 iso"
5. Ubuntu 16.04.1 LTS (Xenial Xerus) - Old Releases <https://old-releases.ubuntu.com/releases/16.04.1/>
6. Select an image → Desktop image → 64-bit PC (AMD64) desktop image to download the Linux OS

Step 3:

Creation of Virtual Machine:

VirtualBox:

1. Open the Virtualbox and click Machine → New. A Virtual Machine Dialog box appears
2. Enter the Name of Virtual Machine and select the version of the machine (32 bit or 64 bit) and click Next
3. Select the type of Hard Disk as Virtual Hard Disk(VHD) and size as default and click Next
4. Select the Main Memory size as default and click Next → Finish.
5. Load the Downloaded Operating System iso file in the CD/DVD (SATA) drive and click Play.

Step 4:

Installation of Guest Operating system:

Ubuntu:

1. Click Play Virtual Machine
2. Click Install Ubuntu → continue → Install Now → Continue
3. Set the time zone and click Continue. Set the keyboard layout , language and click continue
4. Set the Name and password and click continue.
5. Click Restart Now after the OS is installed successfully.

Step 5: Shutdown the guest machine to exit from the Virtual machine.

Conclusion:

Para-virtualization of Linux OS using Oracle's Virtual Box has been implemented.

3. d. Implementation of Para-Virtualization using Oracle's Virtual Box with windows 10 OS

Aim:

To implement of Para-Virtualization using Oracle's Virtual Box with windows 10 OS

Procedure:

Step 1: Installation of Virtual Machine software

VirtualBox:

1. Open Google and type Oracle VirtualBox for Windows
2. Click the link and download the Oracle VirtualBox x86_64.exe
3. After downloading double click the file and click "I Accept" in EULA and install the software **Step 2:**

Windows

1. Open the Google and type **download windows 10 iso**
2. Click the link in and download the versions 64 or 32 bit file

Step 3:

Creation of Virtual Machine:

VirtualBox:

1. Open the Virtualbox and click Machine → New. A Virtual Machine Dialog box appears 2. Enter the Name of Virtual Machine and select the version of the machine (32 bit or 64 bit) and click Next
3. Select the type of Hard Disk as Virtual Hard Disk(VHD) and size as default and click Next
4. Select the Main Memory size as default and click Next → Finish.
5. Load the Downloaded Operating System iso file in the CD/DVD (SATA) drive and click Play.

Step 4:

Installation of Guest Operating system:

Windows:

1. Click Play Virtual Machine
2. After all files are loaded, Click **Install Now** Button.
3. Click **I Accept the terms** in the End User license Agreement.
4. Select the Language and currency and click Next.
5. Select the Drive and Click Next. The Installation process will start.
6. After the installation is complete click Restart Now Button
7. Enter the Username, Password and Time zone.

Step 5: Shutdown the guest machine to exit from the Virtual machine.

Conclusion:

Para-virtualization of windows 10 OS using Oracle's Virtual Box has been implemented.

Program – 4

Installation of Turbo C in Guest OS

4. a. Install C Language in guest OS – Linux and execute the program using VMWare

Aim:

To install C language and execution a C program in Linux guest OS using VMware Workstation.

Procedure:**Step 1:**

1. Download and Install Virtual Machine software – Vmware workstation
2. Download the Ubuntu OS for the Virtual Machine
3. Creation of Virtual Machine – VMware
4. Installation of Linux guest OS - Ubuntu

Step2:**Ubuntu:**

1. Click Play Virtual Machine and login to the Guest OS(ubuntu)
2. Click the icon in top left corner and Type terminal or press Ctrl+Alt+T to open the terminal
3. Type the text editor filename.c to create a empty file with .c extension (Example: nano pali.c)
4. Type the program and save
5. Type gcc filename.c in the terminal and press enter(Example : gcc pali.c)
6. Type ./a.out to execute the program

Step 3:

Write a sample C program and execute it

Coding:

```
# include<stdio.h>
int main()
{
    int A,B, sum=0;
    printf("Enter two numbers A and B:\n");
    scanf("%d %d", &A, &B);
    sum = A+B;
    printf("sum of A and B is:%d",sum);
    return 0;
}
```

Step 4: Shutdown the guest machine to exit from the Virtual machine.**Conclusion:**

Thus, C language has been installed in guest OS – Linux and C program has been executed successfully using VMWare's work station in virtualized environment.

4. b. Install turbo C Language in guest OS – Windows and execute the program using VMware**Aim:**

To install C language and execution a C program in Windows 10 OS using VMware Workstation.

Procedure:**Step 1:**

1. Download and Install Virtual Machine software – Vmware workstation
2. Download the Windows 10 OS for the Virtual Machine
3. Creation of Virtual Machine – VMware
4. Installation of Windows 10

Step 2:

VMware – Windows OS:

1. Click Play Virtual Machine and Install the Windows Operating System.
2. Download the Turbo C++ in the Host System.
3. In the Guest OS Open the File Explorer and select the CD/DVD Drive and right click and select Eject.
4. Shut down the Guest OS.
5. Open the VMware and click Edit Virtual Machine Settings and Select CD/DVD (SATA) and click Browse and select the VMware tools iso file. Click OK
6. Click the Options and select **Shared Folder** and select **Always** in the shared folder settings and click add button and select auto mount option and Map as Network Drive in the shared folder settings and select the Turbo C++ folder. And Click OK
7. Click play virtual machine. After the Guest OS is loaded Enter File Explorer and CD/DVD drive and install the VMware Tools.
8. After Tools are installed the Shared folder selected will be displayed

Step3:

Installation of Turbo C++

1. Open the Turbo C++ folder and double click the setup.exe file to install the Turbo C++ 2. Click “I Accept” in EULA (End User License Agreement) and click Next.
3. Select the destination of Installation and Click Next → Finish
4. Open the Turbo C++ and set the Directories
 - a. C:\\ TURBOC3\\INCLUDE -- Include
 - b. C:\\ TURBOC3\\LIB -- Library
 - c. C:\\ TURBOC3\\ -- Output
 - d. C:\\ TURBOC3\\SOURCE -- Source
5. Create an C program and save in the extension .c and execute the program

Step4:

Write a sample C program and execute it

Coding:

```
#include<stdio.h>
int main()
{
    int A,B, sum=0;
    printf("Enter two numbers A and B:\n");
    scanf("%d %d", &A, &B);
    sum = A+B;
    printf("sum of A and B is:%d",sum);
    return 0;
}
```

Step 5: Shutdown the guest machine to exit from the Virtual machine.

Conclusion:

Thus, Turbo C language in guest OS – Windows 7 has been installed and C program has been executed successfully using VMWare's work station in virtualized environment.

5. c. Install C Language in guest OS – Linux and execute the program using Oracle's Virtual Box

Aim:

To install C language and execution a C program in Linux guest OS using Oracle's Virtual Box.

Procedure:**Step 1:**

1. Download and Install Virtual Machine software – Oracle Virtual Box
2. Download the Ubuntu OS for the Virtual Machine
3. Creation of Virtual Machine – Oracle Virtual Box
4. Installation of Linux guest OS - Ubuntu

Step2:**Ubuntu:**

1. Click Play Virtual Machine and login to the Guest OS(ubuntu)
2. Click the icon in top left corner and Type terminal or press Ctrl+Alt+T to open the terminal
3. Type the text editor filename.c to create a empty file with .c extension (Example: nano pali.c)
4. Type the program and save
5. Type gcc filename.c in the terminal and press enter(Example : gcc pali.c)
6. Type ./a.out to execute the program

Step3:

Write a sample C program and execute it

Coding:

```
# include<stdio.h>
int main()
{
    int A,B, sum=0;
    printf("Enter two numbers A and B:\n");
    scanf("%d %d", &A, &B);
    sum = A+B;
    printf("sum of A and B is:%d",sum);
    return 0;
}
```

Step 4: Shutdown the guest machine to exit from the Virtual machine.**Conclusion:**

Thus, C language has been installed in guest OS – Linux and C program has been executed successfully using Oracle's Virtual Box in virtualized environment.

4. d. Install Turbo C in guest Windows 10 OS and execute C program using Oracle's Virtual Box**Aim:**

To install C language and execution a C program in windows OS using VMware Work station.

Procedure:**Step 1:**

1. Download and Install Virtual Machine software – Oracle Virtual Box
2. Download the Windows 10 OS for the Virtual Machine
3. Creation of Virtual Machine – Oracle Virtual Box
4. Installation of Windows 10 OS

Step2:**VirtualBox – Windows:**

1. Click play button in the virtual machine and load the Windows machine
2. Login to the OS and click Devices→ Insert Guest Addition CD image in guest OS
3. Install VirtualBox Guest Additions and Restart the Guest OS
4. Devices → Shared Folder Settings → Shared Folders → Add New Shared Folder
5. Select other in Folder path and select the directory and folder in host physical memory and click OK.
6. Click the Auto mount check box and click OK → OK. Shared Folder is created.

Step 3: Installation of Turbo C++

7. Open the Turbo C++ folder and double click the setup.exe file to install the Turbo C++.
Click “I Accept” in EULA (End User License Agreement) and click Next.
8. Select the destination of Installation and Click Next → Finish
9. Open the Turbo C++ and set the Directories
 - a. C:\\ TURBOC3\\INCLUDE -- Include
 - b. C:\\ TURBOC3\\LIB -- Library
 - c. C:\\ TURBOC3\\ -- Output
 - d. C:\\ TURBOC3\\SOURCE -- Source
10. Create an C program and save in the extension .c and execute the program

Step4:

Write a sample C program and execute it

Coding:

```
# include<stdio.h>
int main()
{
    int A,B, sum=0;
    printf(“Enter two numbers A and B:\n”);
    scanf(“%d %d”, &A, &B);
    sum = A+B;
    printf(“sum of A and B is:%d”,sum);
    return 0;
}
```

Step 5: Shutdown the guest machine to exit from the Virtual machine.

Conclusion:

Thus, Turbo C language in guest OS – Windows 10 has been installed and C program has been executed successfully using Oracle’s Virtual Box in virtualized environment.

Program - 5

File transfer in Virtual Machine

Aim:

To implement the procedure of file transfer in Client /Server using Virtual Machine using Oracle Virtual Box

Procedure:

- Step 1: Download and install Oracle’s VirtualBox
- Step 2: Create a virtual machine and install Guest OS in virtual machine
- Step 3: In Oracle VirtualBox click → Devices→ Insert Guest Addition CD image
- Step 4: Click Install VirtualBox Guest Additions and Restart the Guest OS

Step 5: Click Devices → Shared Folder Settings → Shared Folders → Add New Shared Folder

Step 6: Select → Other in Folder path → select the directory and folder in host physical memory → OK.

Step 7: Click → Auto mount check box → OK, shared folder will be created.

Step 8: Create a file to be transferred and right click the file and click send to → DV4(Z:)

Step 9: Open the main folder selected and check whether the file is transferred or not.

Step 10: Shutdown the guest machine to exit from the Virtual machine.

1. Authorize → github username and password & click → Authorise AzureAppService
2. Click Azure App Services → Resource → Deployment
3. In Deployment tab enter the following:
 - a. GitHub Action Settings -- Enable
 - b. GitHub account
 - c. Organization -- Same as GitHub Account Name
 - d. Repository -- java-docs-spring-hello-world
 - e. Branch -- main
4. Click Review + Create → Create after entering the above details
5. After Deployment is complete, Click Go to Resource → Overview tab → URL
6. Click the URL link to view the results

Conclusion:

Thus, the procedure of file transfer in client/server in virtual machine using Oracle Virtual Box is implemented